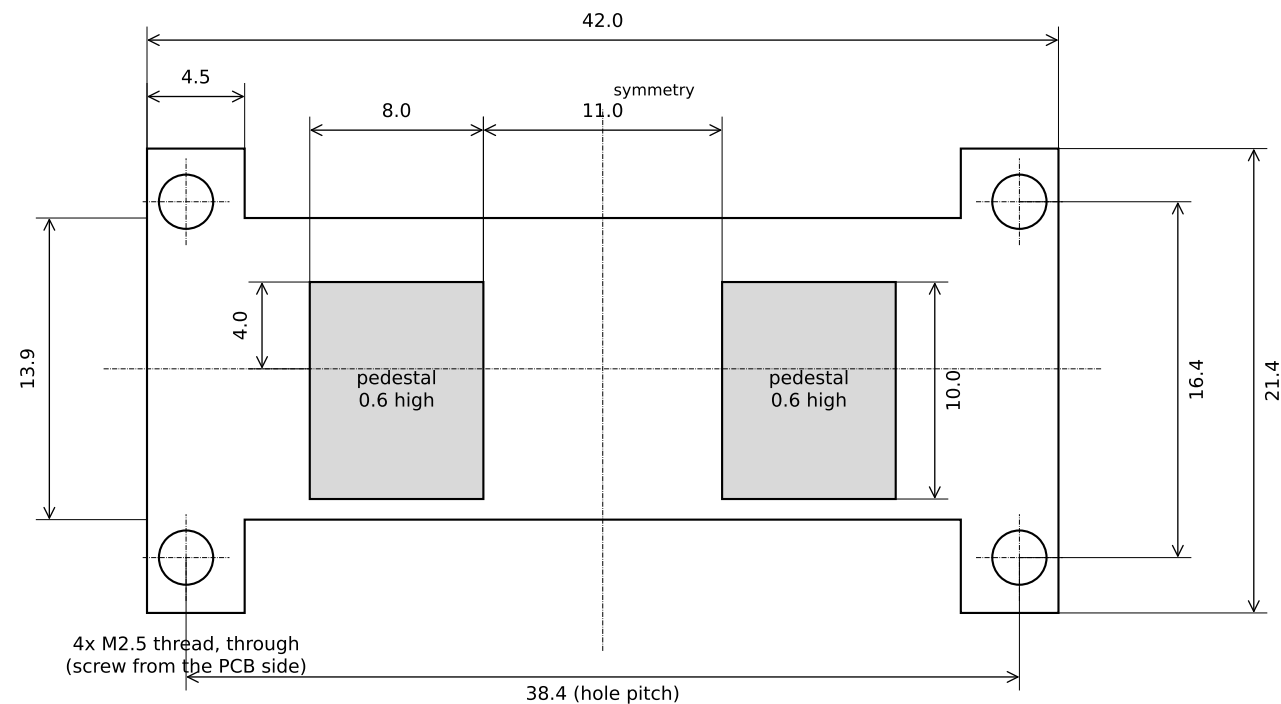
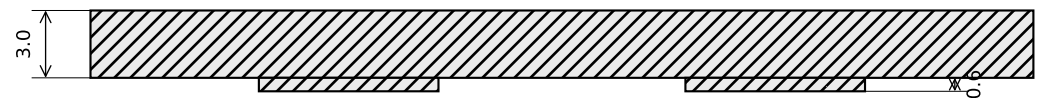


View A: face towards the PCB (pedestals up). Board orientation: driver side at the top.



Section B-B (through the pedestals), heatsink side up



Part	HS1 heat spreader, SPB GaN building block rev A
Material	Aluminium EN AW-6061-T6 (6082-T6 acceptable)
Quantity	1 per board (+1 spare)
Units / tolerance	mm; general ISO 2768-m
Flatness	pedestal faces 0.02 mm, top face 0.05 mm
Pedestal height	0.60 +-0.02, both pedestals equal within 0.02
Surface	pedestal faces and top face Ra <= 0.8, as machined; no anodising (thermal contact)
Edges	break 0.2 x 45 deg, deburr; no burrs on the pedestals
Holes	4x M2.5 thread through, at (+-19.2, -7.7) and (+-19.2, +8.7) from the centre between the cells
Mating	gap pad 8 x 10 mm, 0.5 mm, >= 5 W/mK, insulating, one per pedestal (Bergquist TGP 5000 / Gap Pad 5000S35, 0.020 in, or equivalent)
Assembly stack	PCB / EPC2361 (0.68) / gap pad 0.5 -> 0.4 / pedestal 0.6 / plate 3.0 / thermal compound / LAM 4 K 50 12
Note	EPC2361 top is at source potential: the pad is the only insulation between the cells and the spreader. Check that no parts other than the FETs are higher than 1.6 mm under the plate.
Ref.	scripts/make_3d_models.py (HS_LAM4K5012_assembly), reports/pcb-fab/pcb_fab_report.tex, cooling section

KTH SPB GaN building block

Heat spreader HS1 rev A not to scale, use dimensions
Drawn by: E. Ayaz (generated by scripts/make_spreader_drawing.py)